Swarm Intell
DOI 10.1007/s11721-010-0043-7

Modeling, analysis and simulation of ant-based network routing protocols

Claudio E. Torres · Louis F. Rossi · Jeremy Keffer · Ke Li · Chien-Chung Shen

Received: 1 February 2009 / Accepted: 27 May 2010
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Abstract Using the metaphor of swarm intelligence, ant-based routing protocols deploy control packets that behave like ants to discover and optimize routes between pairs of nodes. These ant-based routing protocols provide an elegant, scalable solution to the routing problem for both wired and mobile ad hoc networks. The routing problem is highly nonlinear because the control packets alter the local routing tables as they are routed through the network. We mathematically map the local rules by which the routing tables are altered to the dynamics of the entire networks. Using dynamical systems theory, we map local protocol rules to full network performance, which helps us understand the impact of protocol parameters on network performance. In this paper, we systematically derive and analyze global models for simple ant-based routing protocols using both pheromone deposition and evaporation. In particular, we develop a stochastic model by modeling the probability density of ants over the network. The model is validated by comparing equilibrium pheromone levels produced by the global analysis to results obtained from simulation studies. We use both a Matlab simulation with ideal communications and a QualNet simulation with realistic communication models. Using these analytic and computational methods, we map out a complete phase diagram of network behavior over a small multipath network. We show the existence of both stable and unstable (inaccessible) routing solutions having varying proper-

C.E. Torres · L.F. Rossi
Department of Mathematical Sciences, University of Delaware, Newark, DE 19716, USA

C.E. Torres
e-mail: torres@math.udel.edu

L.F. Rossi
e-mail: rossi@math.udel.edu

J. Keffer · K. Li · C.-C. Shen (✉)
Department of Computer and Information Sciences, University of Delaware, Newark, DE 19716, USA
e-mail: cshen@cis.udel.edu

J. Keffer
e-mail: keffer@cis.udel.edu

K. Li
e-mail: kli@cis.udel.edu

ties of efficiency and redundancy depending upon the routing parameters. Finally, we apply these techniques to a larger 50-node network and show that the design principles acquired from studying the small model network extend to larger networks.

Keywords Routing protocols · Ant colony optimization · Swarm intelligence · Modeling · Analysis

1 Introduction

Swarm intelligence refers to complex behaviors that arise from very simple individual behaviors and interactions, which are often observed in nature especially among social insects such as ants. Although each individual (an ant) has little intelligence and simply follows basic rules using local information obtained from the environment, such as ant's pheromone trail laying and following behavior, globally optimized behaviors, such as finding a shortest path, emerge when they work collectively as a group (Bonabeau et al. 1999).

Ant-based routing protocols use the metaphor of swarm intelligence to deploy "ants" in the form of control packets to discover routes, reinforce shorter routes via pheromone deposition, and discard longer, less-efficient routes via pheromone evaporation. Throughout this paper, we will use the terms "ant" and "control packet" interchangeably. In addition, the stochastic nature of the ant-based routing protocol allows multiple routes to be discovered and maintained, which provides for a certain degree of fault-tolerance when connections are broken. Prior work has shown that ant-based routing protocols provide an elegant solution to the routing problem of both wired networks (Di Caro and Dorigo 1998) and mobile ad hoc networks (Di Caro et al. 2005; Rajagopalan and Shen 2006; Ducatelle et al. 2010). These ant-based protocols deploy ants under two different circumstances. Proactive ants are sent regularly to discover new routes and reinforce existing shorter routes. Reactive ants are sent in the events of on-demand route discovery and broken routes as can occur in mobile ad hoc networks. In this paper, we will develop our modeling framework for proactive ants on wired networks only, leaving a treatment of reactive ants and mobile ad hoc networks for future work.

The behavior of an ant-based routing protocol is determined by its protocol parameters applied to local protocol rules. We differentiate protocol parameters from network parameters, where the latter include the number of nodes, the size of terrain, etc. The goal of this paper is to study the behavior of an ant-based routing protocol with different protocol parameter values, given a fixed set of network parameters. To achieve this goal, we build an analytic framework for modeling ant-based protocols. Our approach is to use techniques from nonlinear dynamics to understand the evolution of route information for a fixed network (see Strogatz 1994 for a modern text on the subject). To apply these techniques, we model the probability density function of ants across the network nodes and the induced pheromone levels along connections as a nonlinear dynamical system. The resulting nonlinear system captures the evolution of the routing tables. To understand this complex system, we identify stationary points in the resulting system and analyze their stability. Analysis of the nonlinear system is crucial because it provides key insights into the impact of parameters on network performance. While it is highly unlikely that anyone will be able to construct solutions to the nonlinear system on a large network, analysis of stationary points maps local protocol rules to full network performance. In particular, we study small networks that can guide the development of protocols for large networks. In this paper, we use our analytic model to build a phase diagram for a small model network, and then demonstrate its utility in understanding larger, more realistically sized networks.

This paper is organized as follows. In Sect. 2, we position our contribution in the context of other work on the properties of ant-based routing protocols. In Sect. 3, we explain the operations of a typical ant-based routing protocol, including pheromone deposition and evaporation. In Sect. 4, we describe the global modeling and analysis of ant-based routing protocols. In Sect. 5, we validate our modeling and analysis by making direct comparisons to both Matlab simulation with ideal communications and QualNet simulation with realistic communication models. In Sect. 6, we summarize this paper and outline some extensions of the problems solved in this paper.

2 Related work

There have been a variety of contributions in the study of biologically inspired networking algorithms. Yoo, La, and Makowski rigorously studied a simple two router ant-based system with multiple parallel routes (Yoo et al. 2004). The study rigorously determined the long-time asymptotics for the system. This work was augmented by Purkayastha and Baras (2007) who modeled the arrival times of data and control packets along parallel routes between two routers. Similar to the work in this manuscript, the stochastic problem is mapped to a system of ordinary differential equations (ODEs). The authors identify stationary states and analyze their stability. Unfortunately, ant-based routing protocols on realistically sized networks with multiple routers are highly nonlinear by design, and there is little hope that rigorous results classifying the full dynamics of the system will be forthcoming. Nonetheless, essential information can be gleaned from local asymptotic analysis. An analogy can be drawn to systems of nonlinear ODEs where classification of stationary points can provide an essential understanding of the dynamics of the system in cases where an exact solution or a complete understanding of the nonlinear dynamics is not available.

For larger networks, Bean and Costa developed a framework for studying ant-based systems, connecting equilibrium solutions with Wardrop equilibrium, a special case of Nash equilibrium, from traffic flow theory (Bean and Costa 2005). The foundation of Bean and Costa's protocol differs slightly from ours in that they assume that delay between nodes can be measured directly. We assume realistically that nodes possess clocks but not necessarily globally-synchronized clocks so that single-hop delays cannot be measured. Instead, we measure route performance using a simpler but realistically available hop-count. Another difference between the work of Bean and Costa and this modeling effort is that Bean and Costa's routing model follows a succession of unique stationary states. These states correspond to ants following every possible route as if the routing tables were frozen, a valid assumption if the time scale on which ants traverse the network is fast relative to the timescale on which pheromone tables are updated. We propose a fully dynamic model which is more aligned with existing proposed ant-based protocols where the ant flows and the routing data exist in a state space with multiple stationary points, some stable and some unstable. We show that multiple accessible steady states are possible. Also, where Bean and Costa fix the routing exponent to be 2, we construct a detailed phase diagram with the routing exponent as a parameter for a small network, and then demonstrate that the resulting principles apply to larger networks. They propose an off-policy routing scheme as an alternative means of balancing efficiency and network exploration. Others have augmented simulator studies by modeling different aspects of the protocol. Saleem et al. have developed mathematical frameworks for the analysis and measurement of collision probabilities to routing overhead, route optimality and energy consumption (Saleem et al. 2008a, 2008b; Saleem and Farooq 2007). Along similar lines, Zhahid et al. have developed a mathematical framework for

analyzing beehive based protocols (Zahid et al. 2007). Our work is also connected to the work of Roth et al. who analyzed and explored a full range of routing exponents (referred to as "sensitivity parameter" in these papers) to optimize network performance (Roth 2007; Roth and Wicker 2004). His framework is general enough to be exploited for ant-based routing protocols as well as a modified protocol named "termite." In Roth's framework, path utility information updates are included in the model as an external input. In our work, we capture the path utility information update variables by modeling from first principles the transport of ants through the network along with pheromone deposition and evaporation.

3 Ant-based network protocols

Ant-based routing protocols use ant-like control packets and pheromone tables on each node to discover routes between pairs of nodes and to optimize existing routing information. Ant-like control packets can optimize routing tables by reinforcing desirable routes and discarding less desirable routes. Pheromone values determine how ants originating at a source node s and bound for a destination node d will move from one node to the next along a multihop path. The pheromone value τ_{ij} reflects the amount of pheromone on the link from node i to neighboring node j. An ant at node i will hop to node j with probability p_{ij} given by

$$p_{ij} = \frac{(\tau_{ij})^{\beta}}{\sum_{h \in N_i} (\tau_{ih})^{\beta}}, \tag{1}$$

where N_i is the set of all connected neighbors of node i and β is the routing exponent. If $\beta = 0$, routing is random. If β is large ($\beta \to \infty$), routing is deterministic and ants will always select the link with the most pheromone. Our routing function (1) is the simplest possible one that could still be considered ant-based. Other full-fledged protocols might include other quality indicators such as queue length or link quality. As ants hop through the network, they deposit pheromone along links, changing the pheromone values. In addition to deposition, pheromone evaporates over time. Thus, for a route to persist under an ant-based routing protocol, it must be discovered, revisited and reinforced regularly. In the literature, values for β, the deposition rate and the evaporation rate are chosen through experimentation and simulation of network performance. In this manuscript, we develop analytical tools for predicting network performance with different β values without resorting to simulation.

Route discovery and reinforcement in ant-based networks is accomplished when ants deposit pheromone on directed links in the network. Pheromone throughout the network decays through a process designed to mimic the evaporation of pheromone deposited by most ant species:

$$\tau_{ij} \leftarrow \tau_{ij} - \zeta_1 \tau_{ij}, \tag{2}$$

where ζ_1 is a rate constant that will be discussed in the next section. Initially, the network is flooded with ants which discover routes between source-destination pairs. Typically, pheromone values are initialized randomly or uniformly so the first ants find the destination by chance using routes that might be far from optimal. All along the route, the ants maintain a stack of nodes they have visited, a last-in-first-out data structure termed node-visited stack. These ants that are traveling from the source node s seeking a destination node d are called

"forward ants." Once the ant finds the destination node d, it becomes a "backward ant" and retraces its path depositing pheromone along all the directed links:

$$\tau_{ij} \leftarrow \tau_{ij} + \zeta_2 F_{ij}, \tag{3}$$

where F_{ij} is a deposition function and ζ_2 is a rate constant to be discussed in the next section. There is some freedom in choosing deposition functions to accomplish specific purposes, but one regularly discussed in the literature deposits an amount of pheromone inversely proportional to the path cost. One common measure for path cost that we will use throughout this paper is the hop count. This is a natural way to reinforce shorter routes over longer routes.

Features like path cost can be assessed by an individual ant, but there are other desirable features in a network such as fault-tolerance that are properties of the entire network. When existing routes are broken, some ant-based routing protocols deploy reactive ants to discover new routes (Di Caro and Dorigo 1998; Rajagopalan and Shen 2006). The use of reactive ants is one effective strategy for coping with a dynamic network where links are created and destroyed. As we shall see in Sects. 4 and 5, the routing exponent β determines the extent to which single-route or multiple-route paths will be discovered and retained. If the network retains multiple-routes between a source-destination pair, there is redundancy built into the network in the sense that the network remains connected even if a link is destroyed. We investigate fault tolerance by exploring when protocols create multiple route solutions between source-destination pairs. Direct comparisons of the network recovery time when using reactive ants versus built-in redundancy remain a topic for future investigation.

One final comment is that the two protocol processes modeled by (2) and (3) are physically distinct. The evaporation of pheromone requires no communication between nodes, and we expect pheromone to decay precisely as prescribed in (2). The deposition of pheromone (3) requires ants to move between nodes in a lossy and uncertain environment. In such an environment, transmitted ants may not arrive at regular intervals or may disappear entirely when packets are dropped.

4 Global modeling and analysis of ant-based routing protocols

In this section, we derive and analyze global models for simple ant-based routing protocols using evaporation and deposition (see (2) and (3), respectively). We develop a stochastic model by capturing the probability density of ants over the network. If the network consists of m nodes, we define $\mathbf{y}$ to be an m-dimensional vector of probabilities, or a probability distribution. If a single ant were traveling on the network, the kth component of this vector is the probability of finding an ant on the kth node of the network. If there are N ants on the network, then the ith element of $N\mathbf{y}$ is the expected number of ants on the ith node, and $N\mathbf{y}$ is the expected distribution of ants on various nodes of the network. The probability distribution is an evolving quantity which changes with discrete synchronous steps, so that $\mathbf{y}^{(n)}$ is the vector at the nth time step. The ants hop from node to node according to a transition matrix $P^{(n)}(\beta)$ which also evolves in time:

$$\mathbf{y}^{(n+1)} = P^{(n)}(\beta)\mathbf{y}^{(n)}, \tag{4}$$

where the entries of $P^{(n)}(\beta)$ are specified by (1) with one exception: to model the backward ants, we add a single link from d to s with a transition probability of 1 (see Fig. 1). Evaporation is a local process that can be directly mapped from the protocol to the evolution

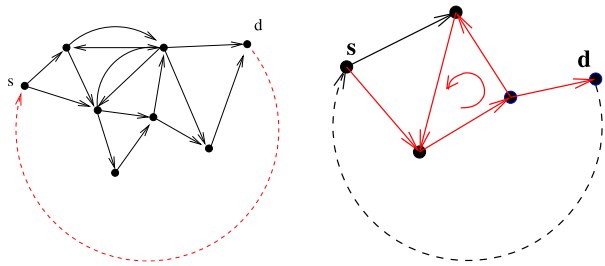


Fig. 1 *Left*: A directed graph expressing the network topology with a link added to create a Markov process. The ant moving from d to s represents an ant retracing its steps and depositing pheromone along its route. *Right*: A network topology which includes an embedded cycle

equation. However, the stochastic events involving ants reversing their paths and depositing pheromone are very complex and are modeled in the expression $F_{ij}^{(n)}$.

In ant-based protocols, ants must be deployed to explore the network, and backward ants will deposit pheromone according to (3). However, the deployment algorithm determines when the ants are released to traverse the network. For the purposes of analyzing the protocol, we need a mathematical model for pheromone deposition which requires knowing how ants are deployed. We propose two different deployment algorithms.

Deployment Algorithm A

1. N ants are released from the source node. The source node resets its clock to $t = 0$.
2. Each ant moves through the network following (1) and maintaining a node-visited stack until it arrives at the destination node.
3. An ant reaching the destination node retraces its steps back to the source. If the ant's route from source to destination is cycle-free (i.e., no node is visited more than once), the ant deposits pheromone (3) along the links it traverses. Otherwise, no pheromone is deposited.
4. When a backward ant arrives at the source, it is destroyed.
5. When the source node clock reaches $t = h_2$, return to step 1.

One variation on step 1 of algorithm A is to release the ants with an offset so that the N ants do not leave simultaneously. This is a useful strategy for avoiding packet collisions and spreading out the communication load over the network.

Deployment Algorithm B

1. N ants are released from the source node.
2. Each ant moves through the network following (1) and maintaining a node-visited stack until it arrives at the destination node.
3. An ant reaching the destination node retraces its steps back to the source. If the ant's route from source to destination is cycle-free (i.e., no node is visited more than once), the ant deposits pheromone (3) along the links it traverses. Otherwise, no pheromone is deposited.
4. When a backward ant reaches the source node, it becomes a forward ant again and proceeds to step 2.

Regardless of how ants are released from the source, we will use the following deposition function:

$$F_{ij} = \frac{1}{N} \frac{1}{H_{sd}^{p}}, \tag{5}$$

where H_{sd} is the number of hops along the route from the source node s to the destination node d, p is an exponent, and N is the number of ants in the network. While it is not always possible to know N precisely, it is possible to estimate N during a simulation. The exponent p is typically taken to be 1 but it can be increased to place a greater emphasis on shorter paths. The factor $1/N$ normalizes the protocol so that performance is independent of the number of ants used. In other words, assuming there is no control packet overhead and that we use a sufficiently large number of ants to discover high quality routes, doubling the number of ants should not significantly alter the pheromone distribution or the routing probabilities. Without the normalization term, using more ants increases the pheromone deposition rate. We note that real protocols do not need to keep track of the number of control packets traversing the network. However, knowing N is essential if we are to use the model to predict network performance. The ways ants are released in the two algorithms will have a substantial impact on how often routes are visited.

4.1 Modeling evaporation

Evaporation occurs on every link all the time, regardless of ant activity. Let us assume that the pheromone level on link ij is updated at discrete, uniformly spaced times $t_1, t_2, \ldots, t_n, \ldots$. Thus, τ_{ij} is a function that is constant on each partition $[0, t_1)$, $[t_1, t_2), \ldots, [t_{n-1}, t_n)$ and so forth, and we define h_1 to be the width of the intervals $t_n - t_{n-1}$. In other words, $\tau_{ij}(t) = \tau_{ij}^n$ if $t \in [t_{n-1}, t_n)$.

We begin with a general evaporation algorithm from (2),

$$\tau_{ij}^{(n+1)} = (1 - \zeta_1)\tau_{ij}^{(n)}, \tag{6}$$

where n is an index identifying particular time intervals and ζ_1 is an adjustable parameter. The evaporation process (6) is applied at regular time intervals of length h_1, and this interval is controlled by the protocol implementation. We would like to understand the protocol mathematically using a model that approximates the system regardless of the specific node hardware and protocol implementation. We can describe the change in the pheromone level from time interval n to interval $n + 1$ as

$$\tau_{ij}^{(n+1)} - \tau_{ij}^{(n)} = -\zeta_1 \tau_{ij}^{(n)}. \tag{7}$$

If we consider a simple model problem where pheromone on a link is decaying steadily without deposition, we know that the pheromone behavior should be consistent, independent of the interval length:

$$\lim_{h_1 \to 0} \frac{\tau_{ij}^{(n+1)} - \tau_{ij}^{(n)}}{h_1} = \lim_{h_1 \to 0} -\frac{\zeta_1}{h_1} \tau_{ij}^{(n)} = f(t), \tag{8}$$

where the limiting function $f(t)$ is not dependent upon h_1. From elementary calculus, we see that $f(t)$ is an approximation to the derivative to τ_{ij} with respect to time. In order to have evaporation, we want this limit to be finite but nonzero, so we require that $\lim_{h_1 \to 0} \frac{\zeta_1}{h_1}$ be a constant which we will call κ_1. Therefore, a dimensionally consistent evaporation algorithm has the form:

$$\tau_{ij} \leftarrow \tau_{ij} - h_1 \kappa_1 \tau_{ij}. \tag{9}$$

Thus, h_1 is determined by the implementation of the protocol, and κ_1 is a tunable parameter that controls the behavior of the algorithm. In the absence of deposition and in the limit as $h_1 \to 0$, we see that pheromone will decay exponentially:

$$\tau_{ij}(t) = Ae^{-\kappa_1 t}, \tag{10}$$

where A is the initial pheromone level $\tau_{ij}(0)$.

4.2 Modeling deposition

Modeling forward ants is very different from modeling backward ants. Forward ants follow a simple Markov process (4) based on the current pheromone values. Backward ants are much more complex because the backward ant uses a stack containing an ordered list of the nodes the ant has visited. The backward ant uses this stack to deterministically reverse its steps, depositing pheromone along the way. In a dynamic network, representing the stack complicates the model considerably because the state of the system needs to include a description of all possible stacks. We remove this complication by modeling the activity of a backward ant as a single step indicated by the dashed link connecting d to s in Fig. 1 (left). In the real protocol, the backward ant visits each ant on the stack in a series of hops as it returns to s, depositing pheromone on each link along the route. A complete model of the full Markov process including ant locations and node-visited stacks would occupy a prohibitively large state space. In our stochastic model, a backward ant will return directly to s in a single hop and in this single step deposit pheromone along the route taken from s to d. For states that are extremely far from equilibrium (e.g., a state where ants are concentrated on a few nodes but probable routes cover a large fraction of the network) this approximation is not valid. However, when the network is close to equilibrium, this simplification is exact. Modeling with deployment algorithm B requires an additional assumption because there is no guarantee of a consistent cohort exploring the network and returning to the source, a feature which is built into algorithm A. To close our model for deployment algorithm B, we assume that at any given time, half of the ants are backward ants and half are forward ants. This assumption is one of convenience to close the model, not one of necessity. In the complete protocol that includes a return stack, if the system is in equilibrium, the number of ants traveling toward the destination on any given route must be equal to the number of ants retracing their steps along this route. (Otherwise, the system would not be in equilibrium.) When we model deposition and apply this assumption, we do so knowing that we will be looking for equilibrium solutions and studying the dynamics near these equilibria where this is a valid approximation.

As a practical matter, ant-based protocols remove ants that revisit a node because a route with cycles is not a good route to use. Unfortunately, this is very difficult to model globally. Instead, we model an idealized protocol where ants that revisit nodes stay alive. (See Fig. 1 (right). Note that there is a cycle in the center.) However, ants that have followed a route with one or more cycles will not deposit pheromone.

For this work, we will model a deposition function F_{ij} where individual ants deposit an amount of pheromone inversely proportional to the distance traveled as described by (2) and (5). The key difference is that the stochastic model captures deposition by having ants at node d deposit pheromone in one step (see dotted links in Fig. 1). In a real implementation, these ants would begin their journey from node d and then retrace their route. In the stochastic model, the ants at node d capture the behavior of all the backward ants when they hop back to s.

To determine ζ_2 in (3), there are two relevant time scales. The first relevant time scale is h_2 which is an input for deployment algorithm A. The second relevant time scale, denoted h_3, is the amount of time required for an ant to make one hop. When analyzing algorithm A, we assume that the time required to move from node to node is much smaller than the time interval over which ants are released into the network, $h_3 \ll h_2$. This guarantees that ants complete their tour before a new cohort of ants is released.

4.2.1 Analysis of deployment algorithm A

For deployment algorithm A, a discrete time interval is the amount of time required for the cohort of N ants to complete their tour to the destination and return. Each ant that followed a cycle-free path will deposit pheromone. Stochastically, (5) will have the form:

$$F_{ij}^{(n)} = \sum_{k=1}^{\infty} \frac{1}{N} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k), \tag{11}$$

where $\tilde{p}_{ij}^{sd}(k)$ is the probability of an ant following a k-hop route from source node s to destination node d passing through link ij without any cycles. Thus, the summation is the expected inverse hop count, $\langle \frac{1}{H_{sd}^p} \rangle$ for a single ant.

The computation or approximation of $\tilde{p}_{ij}^{sd}(k)$ is the most expensive aspect of calculations of this stochastic model. For small networks, this can be done exhaustively by studying each possible route. For larger networks, an algorithm is required. Given $P(\beta)$, there is no known analytic or recurrence relation for computing a cycle-free k-hop path though there are some relatively simple special cases. For instance, the transition matrix for two-hop, cycle-free paths is simply $P(\beta)^2 - \Gamma(P(\beta)^2)$ where $\Gamma(A)$ is the matrix consisting of the diagonal elements of A only. We remain very interested in a general treatment of k-hop cycle-free routes, but lacking such a treatment, we opted for a recursive calculation.

We compute $\langle \frac{1}{H_{sd}^p} \rangle$ recursively by constructing a tree, breadth-first, extending from the source s to the destination d. The tree is constructed by maintaining an ordered list L of nodes visited. At the beginning, this list will consist of only the node s, $L = [s]$. Next, we loop through each neighbor excluding those already in L. We individually add each of these neighbors to the list and apply the recursive function. For instance, if the neighbors of s are nodes 3, 4, and 5, we would call the recursive function with $L = [s, 3]$, $L = [s, 4]$ and $L = [s, 5]$. We repeat this process until d is added to L. If d is added to L, we have found a cycle-free path.

To illustrate this procedure, see Fig. 2. In this example, the aim is to find all the possible paths from node n1 to node n2. On the left of Fig. 2, we have a small example graph, and on the right we have the tree for that graph with all possible paths from n1 to n2. Notice that the legend on each node indicates the number of hops to get to that node from node n1. For instance, n5, 2 hops indicates that to get to node n5 we need 2 hops.

From the tree, we observe that there are 10 arrows leaving the starting node n1, and we have 10 arrows arriving at node n2 at different times. For instance, two paths arrive at n2 in 2 hops. Four paths arrive at n2 in 3 hops. Finally, four paths arrive at n2 in 4 hops. Using this procedure, we build the cycle-free tree.

Although, this procedure seems to be an easy way to solve the cycle-free problem, in practice it becomes computationally prohibitive for larger graphs. For this reason, we define the K-cycle-free tree (Fig. 3). The algorithm is the same as before but we finish the construction of the tree after K hops, which means that K is the maximum number of hops allowed.

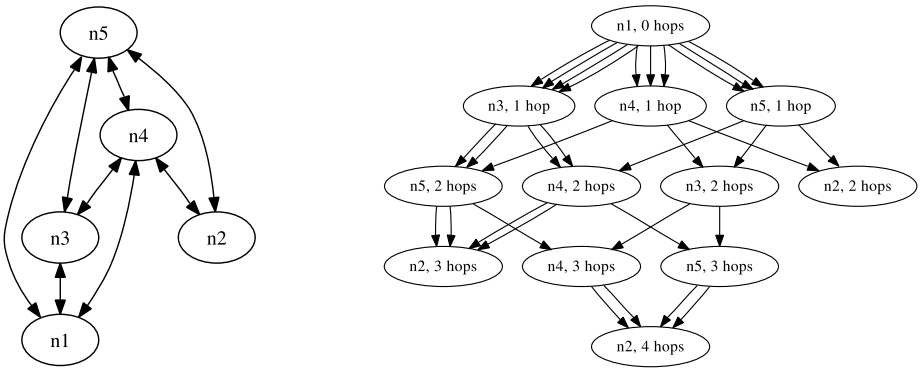
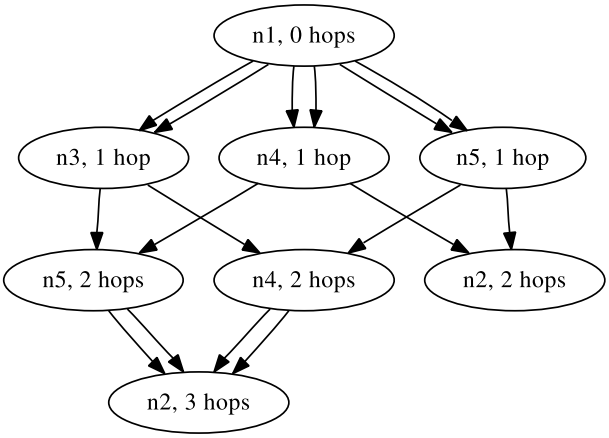


Fig. 2 *Left*: Small example graph. *Right*: Cycle-free tree of small graph on the left

Fig. 3 K-cycle-free tree, with $K = 3$



We point out that the purpose of the K-cycle-free tree is to provide a truncated approximation to the complete problem. As we shall see later, it can be an effective tool for exploring larger networks where an exhaustive search is not possible.

Now that we have a model for constructing cycle free paths, we can model the deposition of pheromone along these paths. As with the process of evaporation, the deposition rate should approach a finite value in the limit as $h_2 \to 0$. For simplicity, we leave out the evaporation terms in the next three equations. The deposition of pheromone by N ants will be

$$\tau_{ij}^{(n+1)} = \tau_{ij}^{(n)} + \zeta_2 N F_{ij}^{(n)}, \tag{12}$$

so we expect that

$$\lim_{h_2 \to 0} \frac{\tau^{(n+1)} - \tau^{(n)}}{h_2} = \lim_{h_2 \to 0} \frac{\zeta_2}{h_2} N F_{ij}^{(n)} = f(t), \tag{13}$$

where $f(t)$ is independent of h_2. From (11), we know that $N F_{ij}^{(n)}$ is independent of h_2, therefore $\lim_{h_2 \to 0} \zeta_2 / h_2$ is a constant which we shall call κ_2. Thus, under algorithm A, combining (11) with the correct time scale for ζ_2, we have the following discrete deposition

model:

$$\tau^{(n+1)} = \tau^{(n)} + h_2\kappa_2 \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k). \tag{14}$$

4.2.2 Analysis of deployment algorithm B

For deployment algorithm B, a discrete time interval consists of a single hop during which all the ants in d move to s and deposit pheromone along the route taken from s to d. In addition to the previously discussed reduction of the backward route to a single link, we make the additional assumption that half of the ants in the network are forward ants and half are backward ants.

If there are N ants in the network, we express the expected number of ants that will become backward ants (modeled as the dashed link in Fig. 1 (left)) as Ny_d. In the true protocol, these ants begin their return home to node s in successive hops. Therefore, the value y_d needs to represent not only the ants starting to reverse their route but in fact all the ants that are reversing their route. Thus, the full population of forward ants is represented by the values of y on all nodes except node d. If we normalize this population and follow the assumption that half of the ants are forward ants, then the expected number of forward ants at node j is $\frac{N}{2}\frac{y_j}{1-y_d}$. The expected number of backward ants beginning their return from d to s at each discrete time step is $\frac{N}{2}\frac{y_d}{1-y_d}$. Therefore, the deposition function has the form:

$$\frac{N}{2}\frac{y_d}{1-y_d} F_{ij}^{(n)} = \frac{1}{2}\frac{y_d}{1-y_d}\kappa_2 \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k). \tag{15}$$

We can use the same reasoning to determine ζ_2 as for deployment algorithm A. In this case, each discrete step corresponds to a single hop, so we have the following deposition function:

$$\tau_{ij}^{(n+1)} = \tau_{ij}^{(n)} + h_3\kappa_2 \frac{1}{2}\frac{y_d}{1-y_d} \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k). \tag{16}$$

Finally, there may be difficulties with studying and using deployment algorithm B consistently in various physical environments. In a realistic network, ants may be dropped and disappear due to congestion or collision. While this presents modeling challenges for both the A and B algorithms, A is self-healing because precisely N new ants are released at regular intervals. With deployment algorithm B, the number of ants still in existence is not known globally. Thus, deployment algorithm A is easier to model and offers more options to protocol designers who need to control the overhead associated with route discovery and optimization. Finally, we have identified three distinct relevant timescales, h_1, h_2, and h_3, in ant-based protocols that affect the dynamics of the full system. In our analysis, we assume that h_1 is small relative to h_2 and h_3. As long as these time scales are small, we shall see in the next section that we can study the dynamics of the system independent of their particular values.

4.3 Linear stability analysis

Once we model deposition, we can understand the global behavior of the protocol by examining stationary distributions of τ and $\mathbf{y}$ and their stability. For these purposes, we shall

assume that $h_1 < h_3 < h_2$. Furthermore, we shall assume that $h_3/h_1 = m_3$ and $h_2/h_1 = m_2$. For deployment algorithm A, this means that each node applies the evaporation algorithm m_2 times while the forward ants complete their tour of the network. For Deployment Algorithm B, this means that each node applies the evaporation algorithm m_3 times while the forward ants complete one hop.

For algorithm A, we shall assume that the pheromone distribution changes a negligible amount during a time period of h_2, and that we can consider the pheromone values frozen while the forward and reverse ants traverse the network. Therefore, we do not need to consider the distribution $\mathbf{y}^{(n)}$ explicitly in our dynamical system:

$$\tau_{ij}^{(n+1)} = (1 - h_1\kappa_1)^{m_2}\tau_{ij}^{(n)} + h_2\kappa_2 \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k). \tag{17}$$

Noting that

$$(1 - h_1\kappa_1)^{m_2} - 1 = -m_2 h_1 \kappa_1 + \frac{1}{2} m_2(m_2 - 1)(h_1\kappa_1)^2 + \cdots, \tag{18}$$

and that $m_2 h_1 = h_2$, we arrive at

$$\tau_{ij}^{(n+1)} = \tau_{ij}^{(n)} + h_2\left(-\kappa_1\tau_{ij}^{(n)} + \kappa_2 \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k)\right) + O\left(h_2^2\right). \tag{19}$$

Alternatively, if m_2 is large, one can refine the discrete model for evaporation by allowing τ to evolve during the m_2 intermediate steps:

$$\tau_{ij}^{(n+1)} = e^{-h_2\kappa_1}\tau_{ij}^{(n)} + h_2\kappa_2 \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k). \tag{20}$$

Unlike algorithm A, in algorithm B, we need to know the distribution of ants on the network. Thus, we have the following system:

$$\mathbf{y}^{(n+1)} = P(\beta)\mathbf{y}^{(n)}, \tag{21a}$$

$$\tau_{ij}^{(n+1)} = (1 - h_1\kappa_1)^{m_3}\tau_{ij}^{(n)} + h_3\kappa_2 \frac{1}{2} \frac{y_d^n}{1 - y_d^n} \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k),$$

$$\tau_{ij}^{(n+1)} = \tau_{ij}^{(n)} + h_3\left(-\kappa_1\tau_{ij}^{(n)} + \kappa_2 \frac{1}{2} \frac{y_d^n}{1 - y_d^n} \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k)\right) + O\left(h_3^2\right). \tag{21b}$$

Both discrete evolution equations (19), (21) can be written in the form:

$$\frac{\tau_{ij}^{(n+1)} - \tau_{ij}^{(n)}}{h} = \mathcal{F}_{ij}\left(\tau^{(n)}\right) + O(h), \tag{22}$$

where h is the time interval between discrete steps and $\mathcal{F}_{ij}$ does not depend upon h. Thus, as $h \to 0$, τ becomes a continuous function of t, and the limit of (22) is

$$\frac{d\tau_{ij}}{dt} = \mathcal{F}_{ij}(\tau). \tag{23}$$

For the remainder of this discussion, we shall discard higher order terms such as $O(h_2^2)$ and $O(h_3^2)$.

Stationary solutions are solutions to (19), (21) that do not change from one time step to the next, so $\tau_{ij}^{(n+1)} = \tau_{ij}^{(n)}$. For Deployment Algorithm A, stationary solutions will have the form:

$$\Lambda \tau_{ij}^{(n)} = \sum_{k=1}^{\infty} \frac{1}{k^p} \tilde{p}_{ij}^{sd}(k), \tag{24}$$

where $\Lambda = \frac{\kappa_1}{\kappa_2}$. Notice that the stationary solutions depend upon β and Λ which we shall designate the *pheromone deposition number*. The pheromone deposition number describes the balance between deposition and evaporation in the network and determines the overall network behavior. For deployment algorithm B, stationary solutions will have the form:

$$\mathbf{y} = P(\beta)\mathbf{y}, \tag{25a}$$

$$\Lambda \tau = \frac{1}{2} \frac{y_d}{1 - y_d} \sum_{k=1}^{\infty} \frac{1}{k} \tilde{p}_{ij}^{sd}(k). \tag{25b}$$

Note that stationary solutions are not necessarily unique for this nonlinear system. In fact, we shall see that simple networks may have many stationary solutions with a fixed β, Λ pair.

The stability of this system can be understood by examining the eigenvalues and eigenvectors of perturbations to (24) and (25). The behavior of small perturbations to (24) and (25) can be classified by studying the eigenvalue/eigenvector pairs associated with the Jacobian matrix $[\frac{\partial \mathcal{F}_{ij}}{\partial \tau_{kl}}]$ evaluated at the stationary solution. If any of the eigenvalues have a positive real part, small perturbations will grow, and the eigenvector provides some information on the "direction" of this growth. If all eigenvalues have negative real part, all perturbations are damped.

5 Validation and discussion

In this section, we study network performance using three complementary methods:

1. A network model of the protocol implemented in Matlab using ideal communications.
2. A network model implemented in QualNet with realistic communication models.
3. Mathematical analysis of the network by studying the stationary states and their stability.

Each method has strengths and weaknesses. The first method is an idealized ant simulation in Matlab independent of the stochastic model. This method removes physical communication effects from the network protocol, so that we can test mathematical results quickly. The second includes more complete physical communication models so that we can explore the limitations of the mathematical theory in the presence of perturbations from propagation delays and other realistic effects. The third method provides a detailed mathematical description of possible equilibrium states and their dependence on protocol parameters.

In the QualNet simulation, we modeled a mesh network of point-to-point links with a data rate of 100 Mbps and a link propagation delay of 1 millisecond. IP was used as the network layer protocol on top of which the ant-based routing protocol operates. Table 1 summarizes the parameter settings used by the ant-based routing protocol in the QualNet simulations.

Table 1 QualNet model parameters

β	0.5, 2
N	2^i ($i = 0$–9), 500
h_1	1 s, 0.01 s
κ_1	0.3, 0.2, 0.4, 0.9
h_2	1 s
κ_2	1, 0.5, 0.25

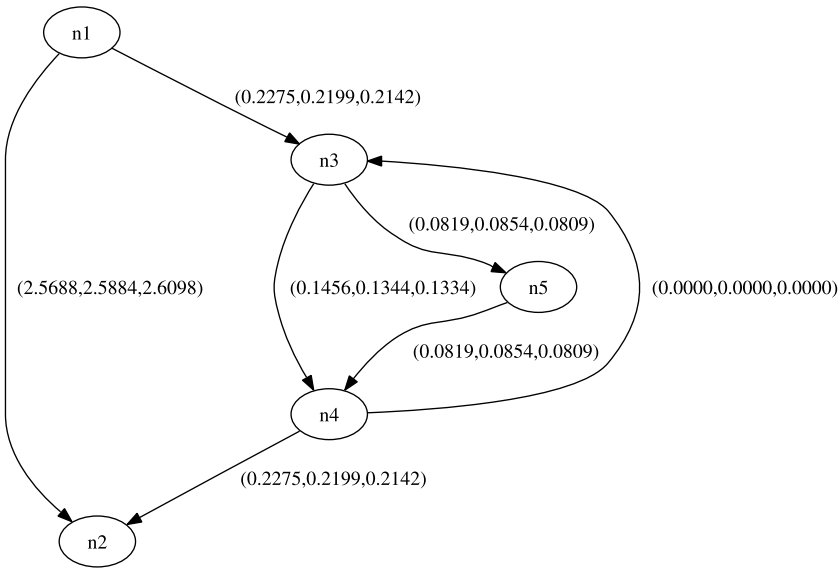


Fig. 4 Comparison of pheromone levels for the stochastic model, the Matlab model with idealized communication and the QualNet implementation of the ant algorithm. This configuration is a stable 3-route solution for $\beta = 0.5$ and $\Lambda = 0.3$. Node 1 is the source node, and node 2 is the destination node. The numbers aside each link represent the model prediction, the Matlab simulation value and the QualNet simulation value, respectively

Once we establish the correspondence between the three methods, we explore network performance using both simulation and asymptotic analysis of the model. First, we study the network behavior to establish design principles on a small, 5 node network where a detailed study is possible. Then, we explore network behavior on a 50 node network to see if these principles apply to larger systems.

5.1 Validation of the three methods

We experimented with both deployment algorithms A and B and found quantitative agreement between the Matlab network model, the QualNet model and the stochastic model on the simple five node network shown in Fig. 4. All the results presented in this section are for algorithm A with $p = 1$. While this is a simple five node network, it has many of the qualities of larger networks including multiple paths from s to d and cycles that can confound the random wanderings of ants. In Fig. 4, we can see a direct comparison of pheromone levels between the stochastic model and the two simulations. For this particular choice of parameters, there is only one stable stationary state. Notice that cycles are possible in the

Table 2 Variances of the pheromone values

N	1	2	4	8	16	32	64	128	256	512
Matlab	0.0155	0.4628	0.3117	0.0670	0.0690	0.0338	0.0149	0.0116	0.0094	0.0087
QualNet	0.7115	0.4467	0.2613	0.1257	0.0481	0.0277	0.0271	0.0101	0.0083	0.0072

graph but both the model and the simulations correctly avoid cycles. The idealized ant simulation uses Matlab's `fsolve` subroutine for solving (24) or (25). The iterative solver for the stochastic model depends upon an initial guess of the solution, and the ant simulations depend upon the initial pheromone distribution. All stable solutions produced using Matlab or QualNet simulations have been verified as a solution to the stochastic model, and all stable solutions produced using the stochastic model can be reproduced using Matlab and QualNet simulations.

To understand the validity of the stochastic model, we performed a study of the variance in pheromone levels in the five-node simulation described in Sects. 5.1 and 5.2 with $\beta = 0.5$ and $\Lambda = 0.3$ when using N discrete ants. This parameter regime has a stable stochastic 3-path solution. These paths have 1 hop, 3 hops, and 5 hops, so maintaining the pheromone table will require enough ants to cover the links in the network. Of course, we do not expect the stochastic model to work well when N is small relative to the number of links. For both the Matlab simulation and the QualNet simulation, we measured the variance:

$$\sigma^2 = \frac{1}{T}\int_0^T \sum_{i,j} \big(\tau_{ij} - \langle \tau_{ij} \rangle\big)^2 \, dt, \tag{26}$$

where $\langle \tau \rangle$ represents the mean value. The summation in i and j is taken over all nodes so that every directed link in the network is included. The variances are shown in Table 2. We see that Matlab and QualNet have similar statistics except when $N = 1$. This is a very special case where the single ant finds a single route. In the course of this QualNet experiment, the ant switched from the single-hop path to a three-hop path briefly which accounts for the very large variation. However, this exception aside, we can see that the stochastic description is valid even for small numbers of ants relative to the number of links. We can also see that nonideal physical effects from QualNet do not hinder the performance of the algorithm.

5.2 Equilibrium states and network protocol performance

In this section, we study network performance in more detail by examining all possible steady states and their stability as we vary β and Λ. In Fig. 5, we provide calculations of solutions to (24). Notice that there are many paths from s to d. Cyclic solutions are possible, but they are not selected by the ant algorithm. When analyzing the stochastic model, we exhaustively construct paths connecting s to d when solving (24), and we intentionally exclude cycles. In the ant simulations, ants which revisit nodes, and hence have cycles in their routes to d, do not deposit pheromone so the simulation and the model are consistent with one another.

Sometimes there are two qualitatively similar, yet distinct, equilibrium solutions. For instance, both S1 and S1p are 3-route equilibria. However, S1 favors the shortest path while S1p favors the longest path. S1 is stable, and S1p is not. Finally, S1 and S1p are not connected to one another in phase space so it is not possible to move from one to the other via a continuous change in parameters.

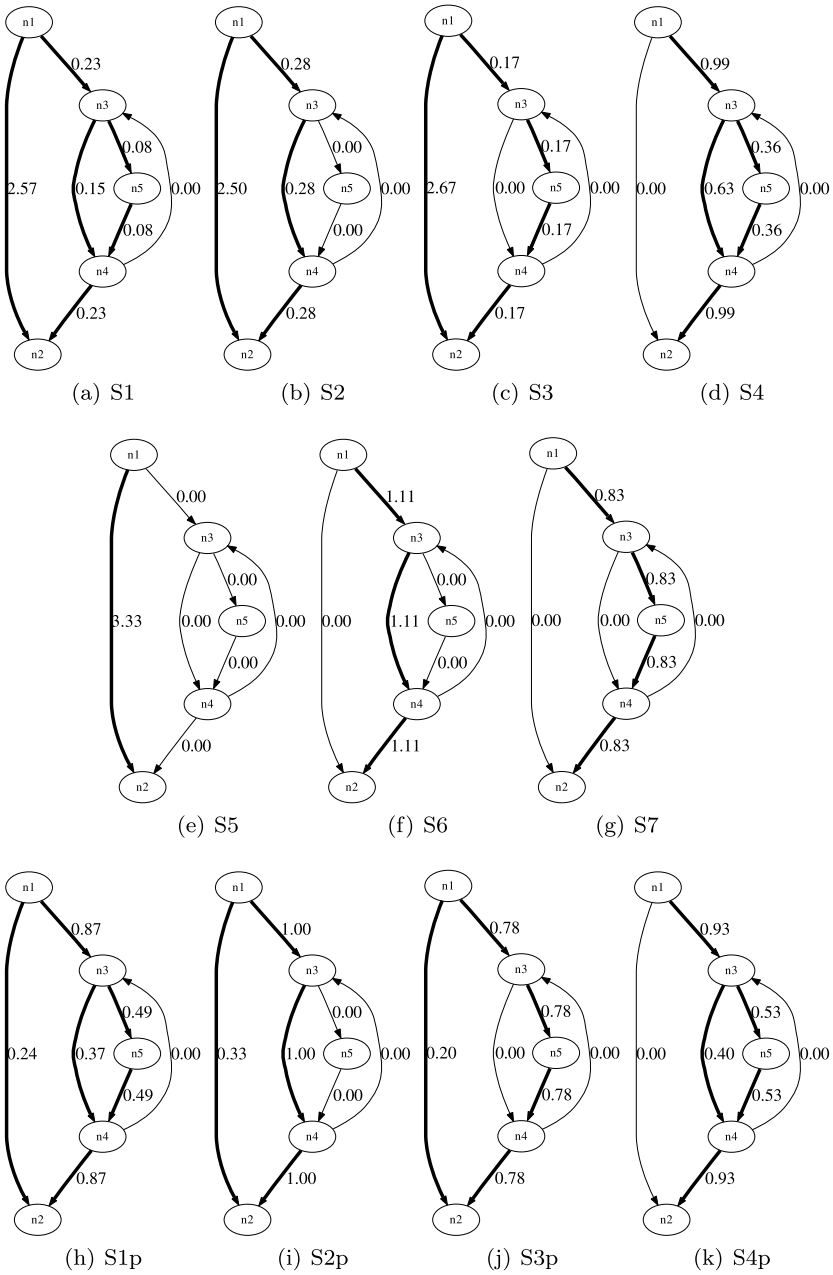


Fig. 5 Potential steady states calculated using the stochastic model. The actual pheromone levels vary depending upon β and κ, but these plots qualitatively represent all possible 3-route (S1, S1p), 2-route (S2–S4, S2p–S4p) and 1-route (S5–S7) solutions. States S1–S7 were calculated with $\beta = 0.5$ and $\Lambda = 0.3$. States S1p–S4p were calculated with $\beta = 2$ and $\Lambda = 0.3$

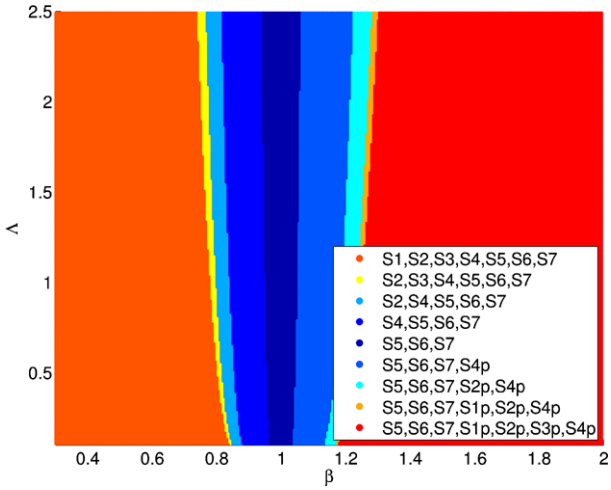


Fig. 6 Equilibrium states as a function of β and Λ

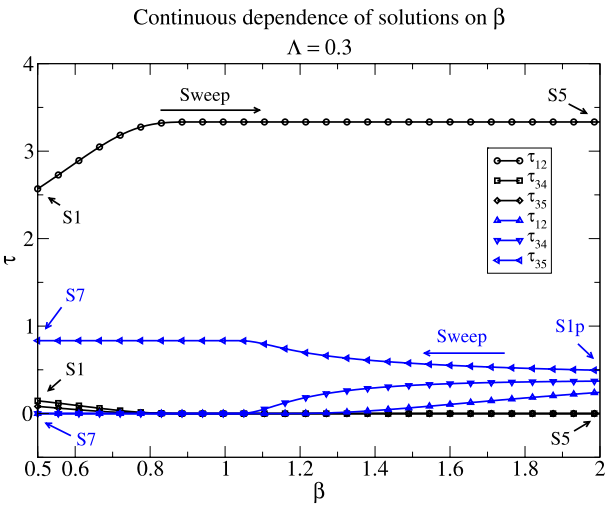


Fig. 7 A demonstration of how the character of solutions varies with β. *Circles*, *squares*, and *diamonds* indicate a sweep from $\beta = 0.5$ to $\beta = 2$ beginning with a multiroute solution S1. Each *curve* represents the amount of pheromone on a specific link in the network. *Triangles up*, *down* and *left* indicate a sweep from $\beta = 2$ down to $\beta = 0.5$ beginning with multiroute solution S1p. All solutions correspond to $\Lambda = 0.3$

To gain further insight into the role of ant algorithm parameters, we present a complete phase diagram of equilibrium states in Fig. 6 (see Fig. 5 for an index of state names). Of course, not all of these states are stable. Notice that 3-route solutions which are the most robust disappear near the critical value of $\beta = 1$. To understand how steady solutions evolve in phase space as a function of β, we continuously varied β beginning with multiroute solutions for $\beta \ll 1$ and $\beta \gg 1$ (see Fig. 7). The stable multiroute solution S1 transitions smoothly to the stable single route solution S5 at $\beta \approx 0.9$. Similarly, the unstable multiroute solution S1p for $\beta \gg 1$ transitions to unstable two-route solution S4p at $\beta \approx 1.15$ and then to the unstable single route solution S7 at $\beta \approx 1$. We did not anticipate that Λ would play a strong role in the number and type of equilibria, and this is confirmed by our analysis. However, Λ does play a role in the network stability because it controls the timescale on which states will move toward stable equilibria.

To understand equilibrium states that are realizable in simulations, we also computed stable equilibrium states in Fig. 8. These correspond to solutions to (24) where all the eigen-

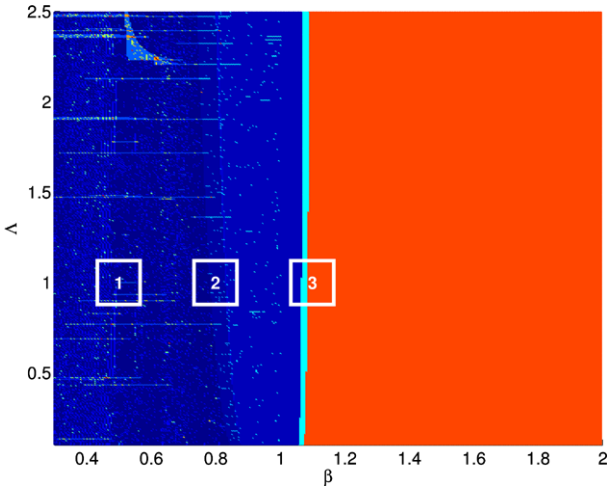


Fig. 8 Stable equilibrium states. In Fig. 9, we expand the square regions labeled 1, 2, and 3, respectively

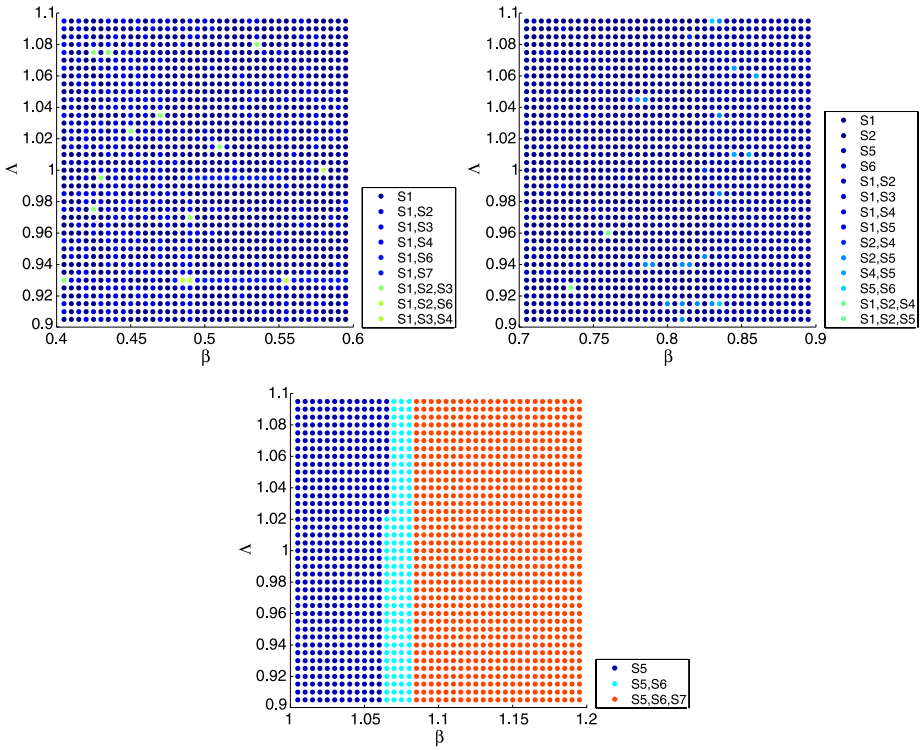


Fig. 9 Expanded view of squares 1 (*upper left*), 2 (*upper right*) and 3 (*bottom*) in Fig. 8

values of the linearized equation have negative real part. For these states, the ant system will damp out small disturbances to pheromone levels. Our calculations indicate a rich structure, shown in detail in Fig. 9, that we are still exploring. Nevertheless, our results show that there are robustness advantages to using $\beta < 1$ because this parameter regime yields stable steady

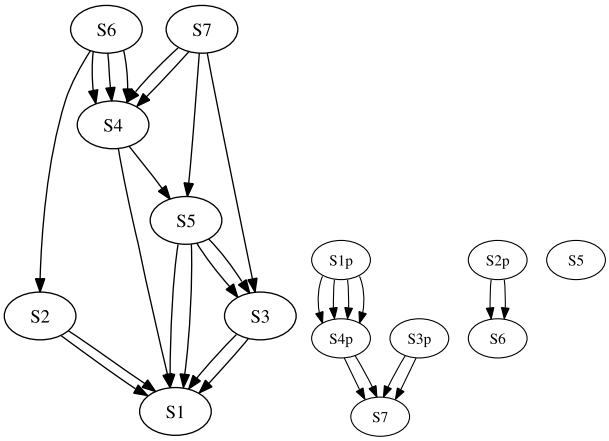


Fig. 10 Evolution of perturbed stationary states for $\beta = 1/2$ (*left*) and $\beta = 2$ (*right*). The diagrams refer back to states in Fig. 5. Each *arrow* represents an unstable eigenvector

states with multiple paths. In published reports on ant protocols, values of $\beta = 2$ are common. Here, we see that values this high would yield a single route solution on the network, but this route might not be the shortest available route. If a link is severed, the protocol must respond with reactive ants rather than relying upon inherent redundancy that is built into stable configurations when $\beta < 1$. The results suggest that values of $\beta > 1$ will lead to more fragile, single-route solutions. Values of $\beta < 1$ will yield naturally redundant configurations.

To understand the mechanisms of instability, we performed finite perturbation analysis on unstable stationary states. For each unstable stationary solution, we added a small disturbance proportional to the unstable eigenvector and then observed the evolution of the system. Based on these findings, we constructed the stability map shown in Fig. 10. We constructed this map by perturbing pheromone stationary states with low-amplitude eigenvectors corresponding to eigenvalues with positive real part. Both positive and negative amplitudes were used. The perturbed pheromone distribution was used as an initial guess for the iterative solver (24) or (25). While the dynamics of the iterative solver are not identical to the dynamics of the stochastic process (19), they have the same residual and produce similar dynamics. Figure 10 shows the evolution of unstable states toward a set of attractors.

5.3 Stability and robustness

We have already seen that the routing exponent has a strong impact on the dynamics and stability of pheromone distributions. The role of Λ is equally important because κ_1 controls the decay timescale. In this case, we measure the relaxation e of the pheromone table for $\beta = 0.5$ back toward the steady state S1 discussed in Sect. 5.2:

$$e = \frac{\|\tau - \tau_{S1}\|_2}{\|\tau_{S1}\|_2}, \tag{27}$$

where τ_{S1} is the stationary solution S1 and $\| \|_2$ denotes the l_2 norm. The initial conditions for τ are given by the perturbed stationary solution $\tau_{S1} + \frac{1}{\Lambda}\delta\tau$ where $\delta\tau$ is a small random matrix. We scale the perturbation by $\frac{1}{\Lambda}$ because τ_{S1} depends upon Λ in (24). Figure 11 demonstrates the predicted relaxation time as we vary Λ. Real networks are constantly perturbed by design. Noise is introduced into the pheromone distribution because N is finite and because of imperfect transmission properties in a real network. It is desirable that a perturbed network relax back to the appropriate stationary solution, and that it does so on

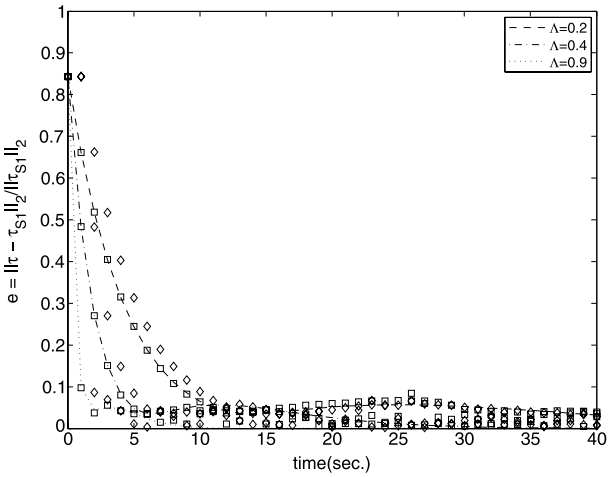


Fig. 11 Relaxation of perturbed pheromone distribution back toward equilibrium. The plot shows the relaxation (27) toward the stationary solution S1. Here, we see that the relaxation time scales with Λ, and that the dynamics of the stochastic model (*lines*), the Matlab simulation (*squares*) and the QualNet simulation (*diamonds*) all agree

a rapid enough timescale to respond to other disturbances. In short, one should choose Λ so that the protocol operates close to equilibrium. Figure 11 also demonstrates the close correspondence between the dynamic stochastic model (19) and the ant-based model (2), (3) implemented both in Matlab and QualNet. The shift in the QualNet time series can be explained by the fact that it does not evaporate pheromone until $t = h_1$ whereas the other dynamic models begin evaporating pheromone at $t = 0$.

5.4 Design principles for larger networks

By exploring and analyzing a small model problem, we infer a design principle. If $\beta < 1$, we anticipate the existence of a collection of stable, multi-path solutions. If $\beta > 1$, we anticipate many possible stable single-route solutions. To test this principle, we generated a random, 50-node network and performed QualNet and Matlab simulations in which we sent ants from one node to another. Simulations were run with 500 ants for 120 seconds (QualNet) until the pheromones settled to a stable value. We performed the same simulation in Matlab. As shown in Fig. 12, values of $\beta < 1$ lead to multiple route solutions whereas values of $\beta > 1$ lead to single route solution using 6 hops. By comparison, when $\beta = 0.5$, there are three 6-hop routes, twenty-eight 7-hop routes and almost two hundred 8-hop routes. While some of these routes have some nodes in common, using values of $\beta < 1$ will yield considerable redundancy with only a slight increase in the expected hop count.

When $\beta = 0.5$, we see that many links remain active in the network. The ants are spread over all the active routes, and we can expect fluctuations in the network. To compare the Matlab and QualNet solutions, we average the pheromone table from $t = 40$ until $t = 120$. In Fig. 13, we see the relative error e_{ij} between average pheromone values:

$$e_{ij} = \frac{|\langle \tau_{ij}^M \rangle - \langle \tau_{ij}^Q \rangle|}{\|\tau^Q\|_\infty}, \tag{28}$$

where $\langle \tau_{ij}^M \rangle$ is the temporal average noted above and $\|\tau^Q\|_\infty$ is the maximum pheromone value over the entire network. A similar measurements of the error for $\beta = 2$ shows that the relative error between Matlab and QualNet solutions is well below 10^{-8}. This is expected because the ants are not distributed across redundant routes in the network.

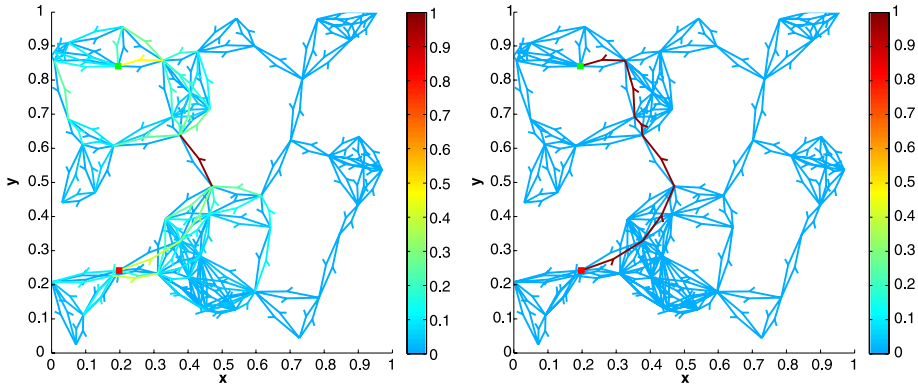


Fig. 12 QualNet simulations of pheromone values for a 50-node network, for $\beta = 0.5$ (*left*) and $\beta = 2$ (*right*) with $\Lambda = 0.3$. Node s is shown as a *square* and node d is shown as a *circle*. Pheromone values are normalized by the maximum pheromone in the network. For network topology, the positions of nodes are represented by normalized coordinates on the x–y plane, and the connectivity of nodes is denoted by the directional links, which also applies to the next two figures

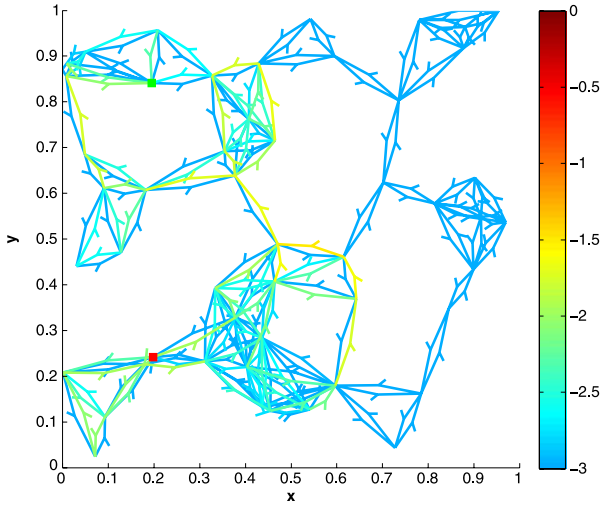


Fig. 13 Relative error between Matlab and QualNet simulations of pheromone values for a 50 node network for $\beta = 0.5$ and $\Lambda = 0.3$. The error shown is the $\log_{10}(e_{ij})$ from (28). Pheromone values are normalized by the maximum pheromone in the network. Links in lightest shade indicate relative errors that are 10^{-3} or less

To gain insight into the differences between QualNet and Matlab simulations, we can also explore instantaneous errors, defined as

$$\frac{|\tau_{ij}^{M} - \tau_{ij}^{Q}|}{\|\tau^{Q}\|_{\infty}}. \tag{29}$$

In Fig. 14, we see that these errors are not uniformly distributed over the network but rather correspond to the strengthening and weakening of individual routes within multiple-route subnetworks. The maximum errors are roughly 10%, but typical errors are closer to 1%, consistent with the controlled study on the 5 node network in Sect. 5.1.

Detailed analysis of steady solutions is also possible for larger systems though it can be expensive, even when using the K-cycle-free approximation for paths connecting the

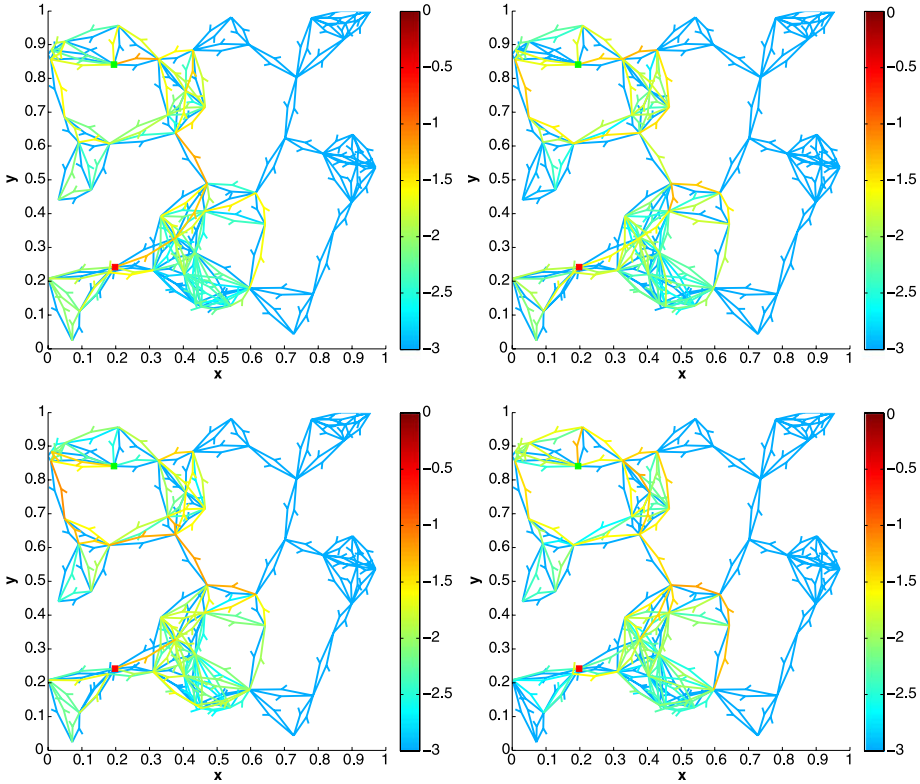


Fig. 14 History of relative error between Matlab and QualNet simulations of pheromone values for a 50 node network for $\beta = 0.5$ and $\Lambda = 0.3$. The error shown is the $\log_{10}(e_{ij})$ from (29). The instantaneous error is shown at times $t = 60$ (*upper left*), $t = 80$ (*upper right*), $t = 100$ (*lower left*) and $t = 120$ (*lower right*)

source and destination nodes. To test the validity of the K-cycle-free approximation on large networks, we used a K-cycle-free approximation for constructing $\tilde{p}_{ij}^{sd}(k)$ and a QualNet solution from Fig. 12 (left) as an initial guess to obtain τ_K by solving (24). We used the relative error metric:

$$e_K = \frac{\|\tau_K - \tau_{\text{QualNet}}\|}{\|\tau_{\text{QualNet}}\|}, \tag{30}$$

where $\|\cdot\|$ is the Frobenius norm to gauge the validity of our approach. In Fig. 15, we see that the error drops toward zero. We do not expect $e_K \to 0$ as $K \to \infty$ because the QualNet simulation results include both realistic communication errors and stochastic noise. The latter effect is a larger scale manifestation of the difference between the continuum stochastic model and the real discrete stochastic system which we explored for the 5-node network in Table 2.

6 Conclusions and future work

In summary, we have modeled ant-based routing protocols, and analyzed critical parameters to extract useful design principles. Our stochastic model successfully captures the stochastic

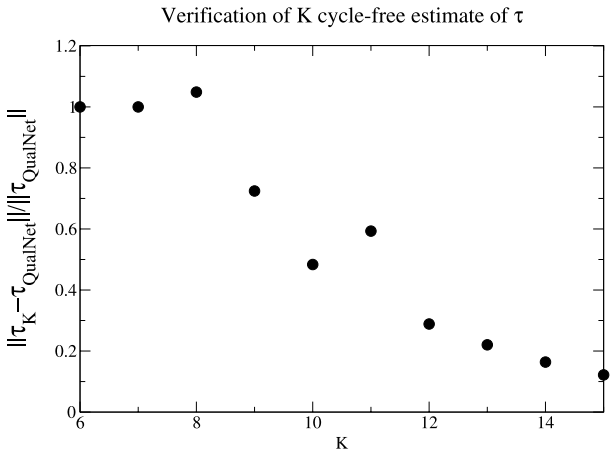


Fig. 15 Error between pheromone levels using exact steady solution using K-cycle-free approximation and using a QualNet simulation

behavior of both a Matlab simulation with ideal communications and a QualNet simulation with realistic communication models. We find that Matlab simulations with no communications overhead and QualNet simulations are self-consistent and consistent with the stochastic model. Our analysis of a simple 5-node network reveals a rich dynamical structure. We see that routing exponents of $\beta < 1$ correspond to the existence of stable, multiroute solutions and unstable single-route solutions. Values of $\beta > 1$ correspond to stable single-route solutions and unstable multi-route solutions. Interestingly, both regions of the phase diagram can be captured with a small number of ants on a network with acceptably low variance. We observe the same behavior on larger networks from which we can make some comments on the routing exponents in general. Ant-based protocols have been successful in part because they are self-healing: broken link will trigger reactive ants to rebuild the pheromone table. However, this work strongly suggests that if $\beta > 1$ this approach can be expensive because few if any alternative routes will be available when the single route breaks. Using a routing exponent $\beta < 1$ causes the ants to build redundancy into the network automatically, so that reactive ants might not be necessary if a link breaks.

The model and methodology described in the paper, although applicable to networks with fixed topology, are a first step toward studies of mobile ad hoc networks. Research is in progress to address the issue of mobility. In mobile applications, we anticipate that the pheromone deposition number Λ will play a more important role because it controls the recovery time for the creation of new links and for the evaporation of discarded links. It is crucial that direct comparisons be made between inherent redundancy when $\beta < 1$ and the use of reactive backward ants when links are broken.

Acknowledgements The authors would like to acknowledge the contributions of Aaron Churchill who worked on this project during the Spring of 2007, the support of Army SBIR grant A072-074-1669, and the helpful comments of our anonymous referees and the editor.

References

Bean, N., & Costa, A. (2005). An analytic modelling approach for network routing algorithms that use "ant-like" mobile agents. *Computer Networks*, *49*(2), 243–268.

Bonabeau, E., Dorigo, M., & Theraulaz, G. (1999). *Swarm intelligence. From natural to artificial systems*. New York: Oxford University Press.

Di Caro, G., & Dorigo, M. (1998). AntNet: distributed stigmergetic control for communications networks. *Journal of Artificial Intelligence Research*, *9*, 317–365.

Di Caro, G., Ducatelle, F., & Gambardella, L.M. (2005). AntHocNet: an adaptive nature-inspired algorithm for routing in mobile ad hoc networks. *European Transactions on Telecommunications, Special Issue on Self-organization in Mobile Networking*, *16*(5), 443–455.

Ducatelle, F., Di Caro, G., & Gambardella, L.M. (2010). Principles and applications of swarm intelligence for adaptive routing in telecommunications networks. *Swarm Intelligence*, *4*(3). doi:10.1007/s11721-010-0040-x.

Purkayastha, P., & Baras, J. S. (2007). Convergence results for ant routing algorithms via stochastic approximation and optimization. In *Proceedings of the 46th IEEE conference on decision and control* (pp. 340–345). New York: IEEE Press.

Rajagopalan, S., & Shen, C. C. (2006). ANSI: a swarm intelligence-based unicast routing protocol for hybrid ad hoc networks. *Journal of System Architecture, Special Issue on Nature Inspired Applied Systems*, *52*(8–9), 485–504.

Roth, M. (2007). The Markovian termite: a soft routing framework. In *Proceedings of the 2007 IEEE swarm intelligence symposium (SIS)*. New York: IEEE Press.

Roth, M., & Wicker, S. (2004). Asymptotic pheromone behavior in swarm intelligent MANETs. In *Proceedings of the sixth IFIP/IEEE international conference on mobile and wireless communication networks* (pp. 335–346). Boston: Springer.

Saleem, M., & Farooq, M. (2007). A framework for empirical evaluation of nature inspired routing protocols for wireless sensor networks. In *Proceedings of the IEEE congress on evolutionary computing* (pp. 751–758). New York: IEEE Press.

Saleem, M., Khayam, S., & Farooq, M. (2008a). Formal modeling of BeeAdHoc: a bio-inspired mobile ad hoc network routing protocol. In *6th international conference on Ant Colony Optimization and Swarm Intelligence* (pp. 315–322). Berlin: Springer.

Saleem, M., Khayam, S., & Farooq, M. (2008b). A formal performance modeling framework for bio-inspired ad hoc routing protocols. In *ACM genetic and evolutionary computation conference (GECCO)* (pp. 103–110). New York: ACM.

Strogatz, S. H. (1994). *Nonlinear dynamics and chaos*. Reading: Perseus Books Publishing.

Yoo, J. H., La, R. J., & Makowski, A. M. (2004). Convergence results for ant routing. In *Conf. info. sci. and systems*. New York: IEEE Press.

Zahid, S., Shahzad, M., Ali, S., & Farooq, M. (2007). A comprehensive formal framework for analyzing the behavior of nature-inspired routing protocols. In *IEEE congress on evolutionary computation, 2007. CEC 2007* (pp. 180–187). New York: IEEE Press. doi:10.1109/CEC.2007.4424470.